THE DESCENT DATUM ALPHA OF A MODULE ON A GROUPOID SCHEME IS AN ISOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition definition-groupoid-module (canonical tag 03LI). cited at proof lines 2879-2880. Supplies the commutative diagram and the axioms making s*e*alpha and t*e*alpha identity.

Source reference. Original definition 03LI: sources/stacks-chow/tags.txt, source lines 4714–4714. Cited at proof lines 2885-2886: its commutative diagram and axioms (e, s, t compositions identity) are pulled back by (i, 1) and (1, i); load-bearing.

Lemma 0.2. *Let S be a scheme, let (U, R, s, t, c) be a groupoid scheme over S. If $(\mathcal{F}, \alpha)$ is a quasi-coherent module on (U, R, s, t, c) then α is an isomorphism.*

Proof. Pull back the commutative diagram of Definition 03LI by the morphism $(i, 1) : R \to R \times_{s,U,t} R$. Then we see that $i^*\alpha \circ \alpha = s^*e^*\alpha$. Pulling back by the morphism $(1, i)$ we obtain the relation $\alpha \circ i^*\alpha = t^*e^*\alpha$. By the second assumption these morphisms are the identity. Hence $i^*\alpha$ is an inverse of α. □

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